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ultra-low power solutions, driving advancements in energy harvesting, like flexible solar cells, to support sustained functionality. However, organic semiconductors still lag in performance due to lower electron mobility and stability concerns, as they are more susceptible to moisture and mechanical stress. Combining soft and traditional silicon components, hybrid designs are emerging to balance flexibility and performance.

Interface engineering, employing techniques like self-assembling monolayers, enhances connectivity with metal electrodes and reduces resistance, while energy-efficient circuits and heat-dissipating substrates tackle power and thermal management challenges critical for wearables. Although compatibility with existing systems hinders IoT adoption, innovations in flexible batteries, antennas, and AI-driven sensors enable applications like wearable healthcare patches for real-time monitoring. The sector's growth depends on ecosystem support, cost reduction through scale, and educating consumers on flexible tech benefits for wider adoption.

Despite its potential, soft semiconductor technology is still in its early stages. Several challenges need to be addressed:

Stability and durability. Organic materials are more prone to degradation than their silicon counterparts. Improving the longevity of soft semiconductors in harsh conditions (such as within the human body or in exposed environments) is crucial.

Efficient charge mobility.

Traditional semiconductors offer high charge mobility, allowing them to operate at high speeds. While organic semiconductors are catching up, they still need to improve, which limits their use in high-speed or

high-power applications.

Manufacturing scale. Producing soft semiconductors on a large scale while maintaining quality and uniformity is complex. New manufacturing techniques, like printing electronics, are promising but still in the developmental stage.

Although challenges in manufacturing and reliability remain, advancements will drive mainstream adoption within five years. During this transition, hybrid devices combining soft and rigid semiconductors will bridge current and future technologies, moving toward fully flexible systems. The drive for sustainability also fuels development in biodegradable materials to reduce electronic waste.

Innovations and challenges on the road ahead

Researchers are exploring new materials, such as bio-inspired substances that mimic nature's self-healing abilities, and nanomaterials like graphene, to improve soft semiconductor performance. These developments open doors for applications in soft robotics, smart textiles, and flexible displays, while collaborations between industry and academia aim to address remaining challenges in durability, performance, and scalability.

Companies like DuPont, 3M, MC10, and LG Display are leading innovation in flexible electronics, health applications, and materials science. As the technology matures, it promises to bring products that enhance comfort, resilience, and functionality in ways traditional devices cannot. **EFY**

Akanksha Sondhi Gaur, a senior journalist at EFY, has a robust blend of seven years of industrial and academic prowess. Passionate about electronics, she has penned numerous research papers and owns a German patent.